VIRGINIA COMMONWEALTH UNIVERSITY

Statistical Analysis and Modelling (SCMA 632)

A2: Analysing IPL Performance and Player Salaries: A Statistical Approach

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CONTENTS

Introduction

This report dives into the world of IPL cricket using data. We're looking at how players perform and how much they get paid. We've got detailed ball-by-ball match data and player salaries up to the 2024 season. Our main goal is to figure out who the best players are, understand the patterns in their performance using statistics, and see if there's a clear link between how well they play and their salary. A big part of this work also involves comparing how we do all this analysis using two different computer programs: Python and R. This way, we'll not only show what we found, but also point out any differences in how these tools handle the data.

Objectives

- Find the top 3 batsmen (by total runs) and top 3 bowlers (by total wickets) for every IPL season using our performance data.
- Figure out the best statistical pattern (called a "probability distribution") that describes how a player scores runs or takes wickets in a match. We'll use special tests to check how good these fits are.
- See if there's a connection (a "correlation") between how many runs batsmen score or wickets bowlers take, and their salaries for the 2024 IPL season.
- Compare our findings from Python and R. We'll specifically look at whether the salary-performance links are similar and if both programs pick the same "best-fitting" statistical patterns for player performance.

Business Significance

- **Picking Players & Building Teams**: Knowing how players perform over time and predicting their output helps IPL teams make smarter choices when buying players in auctions and deciding who plays in each game. This means less risk and better team setups.
- **Justifying Salaries**: By matching player performance to their pay, team managers can see if players are getting what they're worth. This helps them confirm current salaries or change them for future contracts, making things fair.
- **Predicting Game Outcomes & Strategy**: If we know the likely range of runs a batsman might score or wickets a bowler might take, teams can use this for planning. It helps them decide game strategy, understand risks, and can even be useful for fantasy cricket games and betting.

Results

- Top Players Each Season
    - Looking at the *IPL_ball_by_ball_updated_till_2024.csv* file, we consistently found the top players across all seasons. For the 2024 season, here are the best ones:
        - Top 3 Batsmen (Total Runs)
            - RD Gaikwad – 509 Runs
            - V Kohli – 500 Runs
            - B Sai Sudharsan – 418 Runs
        - Top 3 Bowlers (Total Wickets)
            - HV Patel – 19 Wickets
            - Mukesh Kumar – 15 Wickets
            - Arshdeep Singh – 14 Wickets

- How Salary Connects to Performance (2024 Season)
    - To see if salary and performance go together, we linked player names from the *IPL SALARIES 2024.xlsx* file with their stats. Here's what we got for the "Pearson Correlation":
        - Salary vs. Runs (Batsmen)
            - Python – 0.306
            - R – 0.453
        - Salary vs. Wickets (Bowlers)
            - Python – 0.057
            - R – 0.214
- Best Statistical Patterns for Player Performance
    - For Batsman N Pooran (runs per match)
        - Python – *dgamma* was best fit with p-value of 0.1738 with a K-S test.
        - R – *gamma* was best fit with p-value of 0.03335 with a K-S test.
        - Conclusion: This shows that Python and R picked different best patterns and gave different P-values. This can happen because they might have looked at a different set of patterns, used different math to fit them, or even handled tricky data (like ties) in slightly different ways.
    - For Other Top 3 Players
        - 2024 Batsmen
            - RD Gaikwad:
                - Python – *nct*
                - R – *lnorm*
            - V Kohli:
                - Python – *beta*
                - R – *exp*
            - B Sai Sudharsan:
                - Python – *f*
                - R – *exp*
        - 2024 Bowlers

- HV Patel:
    - Python – *alpha*
    - R – N/A (No good fit found)
- Mukesh Kumar:
    - Python – *alpha*
    - R – N/A (No good fit found)
- Arshdeep Singh
    - Python – *t*
    - R – N/A (No good fit found)

- Conclusion: R often couldn't find a good fit for bowlers. This might be because bowlers often take 0 wickets in a match (which can make it hard for some patterns to fit), or there might be a small issue in how R's *get_best_fit* function picks the "best" pattern. It looked like R might have accidentally picked the pattern that was a *worse* fit sometimes. This really highlights how important it is to double-check your code logic carefully!

Interpretations

Both Python and R agreed on who the top batsmen and bowlers were over the seasons. This means our basic way of counting runs and wickets is solid. Players like RD Gaikwad, V Kohli, and HV Patel showing up consistently at the top tells us they're super important to their teams. They're likely the backbone of successful IPL squads.

The link between a player's salary and their performance (runs or wickets) in 2024 is positive, but not super strong. High-paid batsmen generally score more runs, but there's more to their value than just runs. Things like how fast they score, how consistent they are under pressure, if they lead the team, or even their popularity, probably play a big role in their salary. For bowlers, the link between salary and wickets is even weaker. This suggests that bowlers might get paid more for things like not giving away too many runs, taking wickets at key moments (like in the last few overs), their reputation, or special skills (like a unique spin).

The differences we saw in correlation numbers (like 0.306 vs. 0.453 for batsmen) and especially in the "best-fitting" statistical patterns are a big deal. These differences probably come from a few things:

- **How Names Are Matched:** Python uses *fuzzywuzzy* and R uses *stringdist* for matching player names. Even if we set them up similarly, they work a bit differently inside. This means they might have matched a slightly different group of players between the salary and performance data, which then changes the correlation results.
- **How Statistical Patterns Are Found:** Python's *scipy.stats.dist.fit()* is a general tool for finding these patterns. R's *fitdistrplus* package does a similar job, but they might use different math behind the scenes to get the best fit. Plus, one program might have more special patterns available than the other.
- **Checking How Good the Fit Is:** Both analyses used the K-S test, but how they decided which pattern was "best" from a list of options might have been different. For example, we noticed a potential issue in R's *get_best_fit* function where it might have picked the pattern that was *worst* at fitting the data (by choosing the highest K-S statistic, which is bad) instead of the best. This could explain why R sometimes said "N/A" for bowlers and generally had different results from Python.
- **Data Quirks:** Cricket data, especially for wickets, often has a lot of "zeros" (no wickets taken in a match). Different programs or how we use them might handle these zeros in various ways, which can mess with how well a pattern fits.

So, while both R and Python are super powerful tools, their different libraries, hidden math, and small default settings can lead to different answers. This teaches us that even when you try to do the "same analysis," the results can vary if you don't fully understand how each tool works behind the scenes.

Recommendations

- Make Name Matching Consistent: To get more reliable correlation numbers, we should create a super clear way to match player names across all data. Maybe we can try more advanced matching tricks or manually check important player names. If we had unique player IDs, that would be even better!
- Dig Deeper into Salary Reasons: Since just runs and wickets don't fully explain salaries, we should investigate other things that make a player valuable. Like:
  - **Long-Term Performance**: How good they've been over many seasons, not just one.
  - **Player Roles**: Are they a power-hitter, a death bowler? Look at stats that matter most for their specific job.
  - **Auction Fun**: How do teams bid? How much demand is there for a certain type of player?
  - **Hidden Talents**: Do they captain the team well? Are they amazing at fielding? Can they do multiple things (bat and bowl)?
- **Improve How We Find Statistical Patterns**
  - **More Patterns**: Make sure we're trying out a wide range of statistical patterns in both R and Python that make sense for cricket stats, especially for data that has a lot of zeros.
  - **Smarter "Best Fit" Choice**: We need to clearly decide how to pick the "best" pattern in a smart way. And fix that issue in the R *get_best_fit* function.
  - **Test on New Data:** We should test our chosen patterns on data they haven't seen before to make sure they're truly good.
- **Break Down Player Types:** To get super precise insights, we should group players by what they do (e.g., top batsmen, middle-order batsmen, fast bowlers, spin bowlers). Then, we can do the performance and salary analysis for each group, as patterns might be different for each.
- **Look at Performance Over Time:** We should study how players' performance and salaries change over many seasons. This helps us understand their careers, when they play their best, and if teams are getting good value for their money in the long run.

References

- Data provided through Canvas Portal.